

INES KRAIM AI & Data science engineer

 ineskraim22@gmail.com  92930425  Ariana, Tunis  Portfolio  LinkedIn  GitHub

PROFILE

AI & Data Science engineer specialized in multi-agent systems, LLM orchestration, and graph-based intelligence. Built an enterprise AI platform ranked Best FYP Project by PwC, combining RAG, knowledge graphs, and GNNs to automate business workflows. Strong foundation in machine learning, deep learning, and end-to-end data pipelines, from model design to production deployment.

PROFESSIONAL EXPERIENCE

- 02/2026 – 05/2026
TALAN Tunisie Consulting

AI Engineering Intern , (Enterprise AI Assistant), Ranked Best FYP Project by PwC
 - Designed a multi-agent AI platform (FastAPI + React) combining an internal enterprise assistant with an external market intelligence system
 - Built a conversational assistant with LangGraph orchestrating HR, CRM, and ERP systems, enabling natural language queries, RAG-based document retrieval, and automated workflows (reports, emails)
 - Modeled the organization as a Neo4j knowledge graph integrated with PostgreSQL, powering role-aware dashboards that adapted access and responses by user profile
 - Built a real-time market analysis pipeline using an LLM-based analysis layer and a temporal knowledge graph, applying a Graph Neural Network (TGAT) to model event dependencies and estimate business impact
 - Extended the system with a competitive intelligence module (web scraping + LLM analysis) to track competitors and surface business opportunities/threats
- 06/2025 – 08/2025
Zaitouna Bank

AI Developer Intern , (OCR & LLM Document Processing)
 - Developed a Django web app automating receipt extraction and accounting workflows.
 - Integrated a multi-OCR pipeline (Doctr, Tesseract, Docling) for accurate text recognition.
 - Applied LLMs (Gemini, Qwen, Ollama for fallback) to structure and correct OCR outputs.
 - Built a real-time analytics dashboard and automated Excel report exports.
 - Improved data accuracy and reduced manual entry time through AI-driven automation.
- 06/2024
National Council of Statistics (CNS)

Data Science Intern , (PCA & Data Visualization)
 - Performed data cleaning and preparation on demographic and socioeconomic datasets from Excel files.
 - Applied PCA to reduce 61 initial variables to 14 key components revealing the main axes of regional variation.
 - Developed an interactive R Shiny application featuring a choropleth map of Tunisia and dynamic visualizations, enabling users to explore the distribution of 14 variables across governorates, districts, and offices.

PROJECTS

- Medical Image Analysis – Binary Classification** , 1st Place, Bootcamp
 - Developed a deep learning model for early melanoma detection using the ISIC 2024 dataset
 - Fused dermoscopic images and patient metadata through a dual-input architecture (EfficientNetV2 + Dense layers)
 - Designed an end-to-end pipeline: data balancing, data augmentation, transfer learning, and patient-level stratified splitting
 - Trained the model using Adam optimizer and Binary Cross-Entropy loss**Technologies:** Python, TensorFlow, KerasCV, OpenCV, Pandas, NumPy, scikit-learn
- Automated Trading with Reinforcement Learning** 
 - Developed an intelligent trading agent based on the A2C (Advantage Actor-Critic) algorithm
 - Trained the agent on real financial time series data (GME stock) using Gym environments
 - Optimized a stochastic policy to maximize net profits while incorporating risk penalties (drawdown control)
 - Achieved strong performance: +18.73% return, Sharpe ratio = 1.42, outperforming traditional strategies**Technologies:** Python, Stable-Baselines3, Pandas, NumPy, Matplotlib, Gym

EDUCATION

- 2023 – 2026

Higher School of Statistics and Information Analysis (ESSAI), Engineering Cycle – Data Science
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Mining, Big Data, Business Intelligence, Statistical Inference, Bayesian Statistics, Time Series Analysis, Data Analysis, Operations Research, Econometrics, Database Systems
- 2021 – 2022

Preparatory Institute for Engineering Studies of Tunis, Preparatory Classes – Mathematics/Physics

SKILLS

- Programming & Development**
Python (pandas, NumPy, scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face Transformers), R, SAS, C, Java, SQL, HTML/CSS/JS, Django, FastAPI, React, PostgreSQL, Neo4j, Docker, REST APIs

Machine Learning & IA
Supervised & Unsupervised Learning, Deep Learning (CNN, RNN, LSTM), Reinforcement Learning, Graph Neural Networks (GNN), NLP, LLMs, RAG, Vector Databases, LangChain, LangGraph, CrewAI, AutoGen, Fine-Tuning (LoRA/QLoRA), BERT, Prompt Engineering, Text Classification
- Development Tools, Project Management & Design**
Git/GitHub, VS Code, Jupyter Notebook, Power BI, Tableau, Matplotlib/Seaborn, Draw.io

LANGUAGES

- English**
Proficient

French
DEL F B2

Arabic
Native

ASSOCIATIVE LIFE

- 2024

Vice-President, Enactus ESSAI: Co-led a social entrepreneurship project on early lung disease detection, advancing past the early-stage round of the competition.
- 2023

HR Manager, Machine Learning Club : Coordinated weekly ML/DL workshops and hands-on project sessions for students.